

Autodesk® **Scaleform**®

Scaleform CLIK

Scaleform CLIK

Prasad Silva, Matthew Doyle

2.0

2010 8 19

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Autodesk® Scaleform® 4.2

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How to Contact Autodesk Scaleform:

Document	Scaleform 4.0 CLIK AS3 User Guide
Address	Autodesk Scaleform Corporation 6305 Ivy Lane, Suite 310 Greenbelt, MD 20770, USA
Website	www.scaleform.com
Email	info@scaleform.com
Direct	(301) 446-3200
Fax	(301) 446-3199

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1

Scaleform®
CLIK™)
[CLIK](#)
CLIK

Common Lightweight Interface Kit
CLIK
Scaleform
CLIK

[CLIK](#)

1.1

Scaleform CLIK
4.0

ActionScript™ 2.0 (AS2)
PC
CPU
CLIK 15

Scaleform
UI

1.1.1. Scaleform CLIK

UI

Button	Slider
ButtonBar	StatusIndicator
CheckBox	ProgressBar
RadioButton	UILoader
Label	ScrollingList
TextInput	TileList
TextArea	DropDownMenu
ScrollIndicator	Dialog
ScrollBar	NumericStepper

API

InputManager	Locale
FocusManager	Tween
DragManager	DataProvider
PopUpManager	IDataProvider
EventDispatcher	IList
Delegate	IListItemRenderer

Constraints	GameDataProvider
-------------	------------------

CLIK
CLIK
CLIK API
CLIK
CLIK Flash
CLIK

1.2

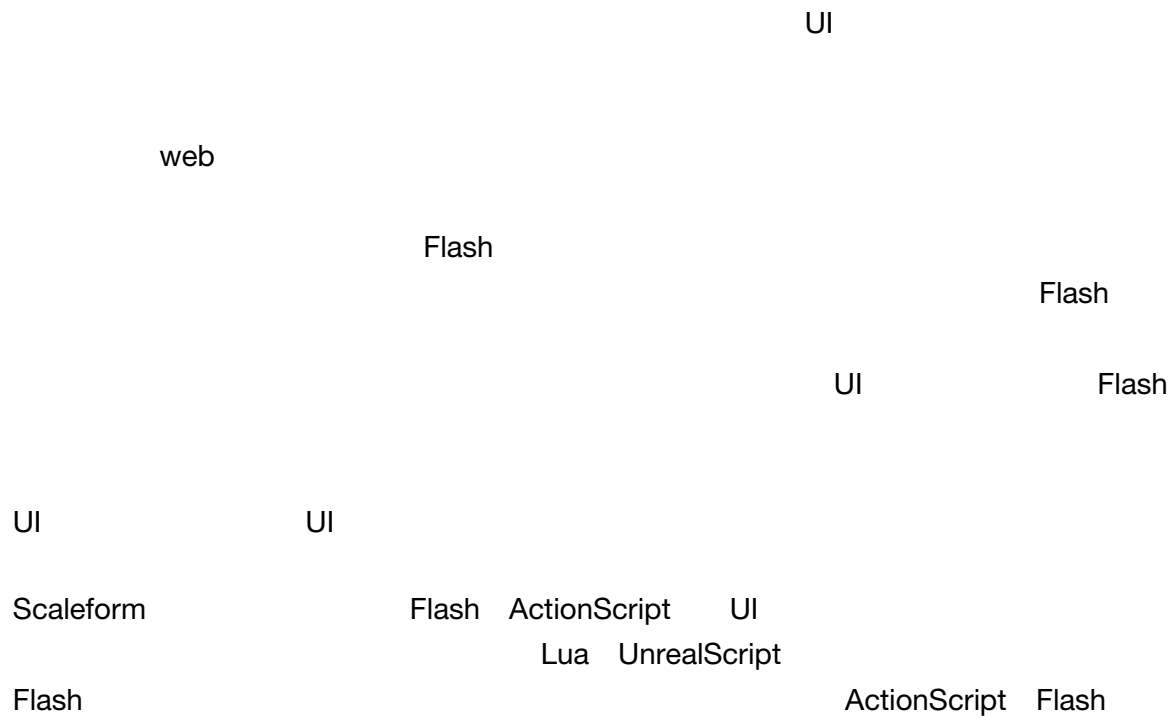
Flash “ ” Scaleform
 CLIK Scaleform gskinner.com

gskinner.com	Grant Skinner	Grant Skinner	Adobe	Flash Creative Suite® 4 (CS4)
		Flash		gskinner.com
		http://gskinner.com/blog		

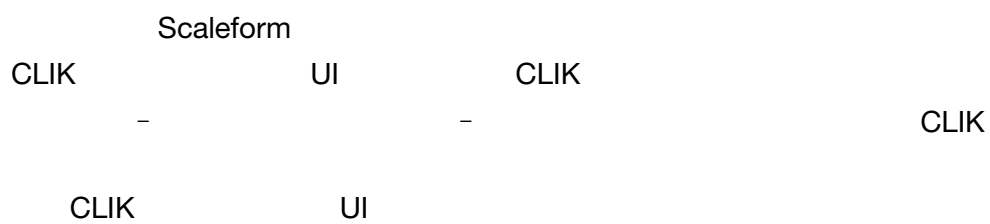
CLIK Flash studio CLIK
*C:\Program Files\Scaleform\Scaleform
 4.2\Resources\AS2\CLIK\components\CLIK_Components fla*

1.3 UI

UI Microsoft
 Visio® Flash UI
 Scaleform CLIK
 UI Flash Visio
 Flash



2

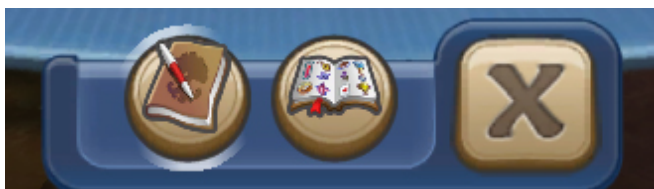


- **User interaction**
- **Component setup** Flash
- **States**

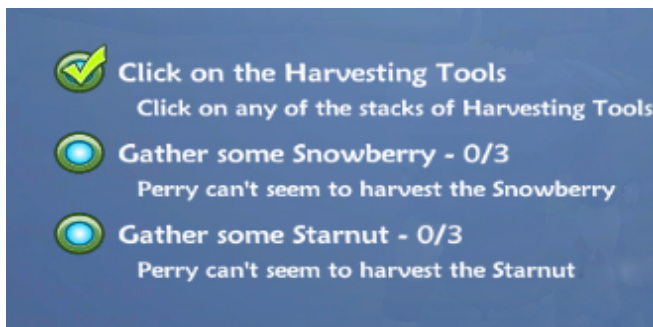
- **Inspectable properties** Flash
- **Events** UI
- **Tips and tricks**

2.1

Button CheckBox Label TextInput
 TextArea Button CheckBox Button Label
 TextInput TextArea



1 **Free Realms**



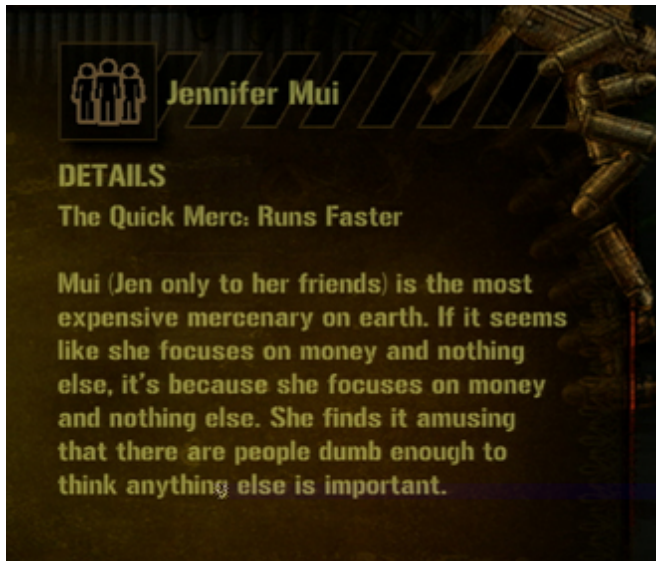
2 **Free Realms** Check box



3 **Free Realms** Label



4: **Crysis Warhead** Text input



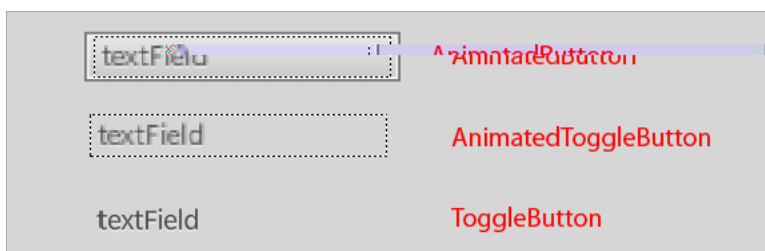
5 *Mercenaries 2* Text area

2.1.1 Button



6

Button CLIK But
 ton gfx.controls.Button textField ScrollBar Slider



7 **AnimatedButton** **AnimatedToggleButton** **ToggleButton**

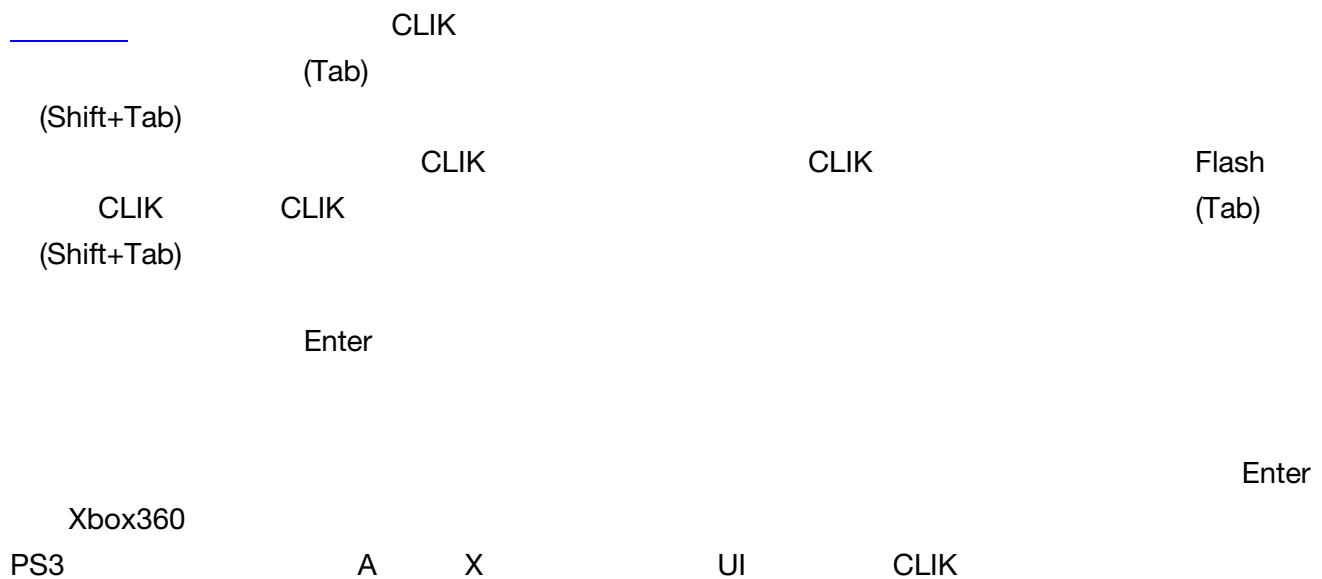
CLIK
Button

ToggleButton AnimatedButton

AnimatedToggleButton Button.fla

2.1.1.1

Programming



2.1.1.2

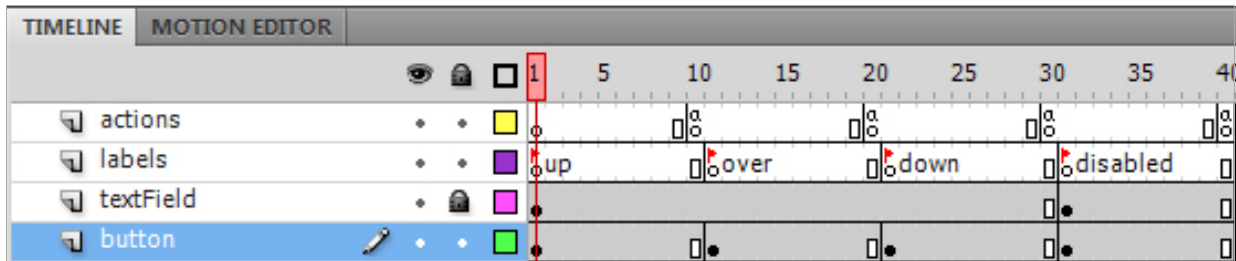
- CLIK Button MovieClip
- **textField:** () TextField
 - **focusIndicator:** () MovieClip
- MovieClip
“ show ” “ hide ” over
- Button
- over

2.1.1.3

CLIK

- **up**

- *over*
- *down*
- *disabled*



8 Button

Flash

[CLIK](#)

2.1.1.4

Flash IDE Component

Inspector

Parameters

CS4

Window

Component Inspector

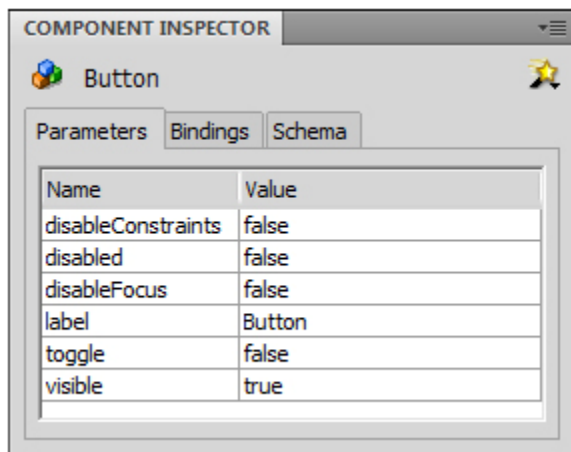
(Shift+F7)

Component

Inspector

“ ”

AS



9 CS4

:

disableConstraints	Button	textField
	true	

rollout	<i>mouseIndex</i>	0-3	
Press	<i>mouseIndex</i>	0-3	-
doubleClick	<i>mouseIndex</i>	<i>doubleClickEnabled</i>	true
	<i>mouseIndex</i>	0-3	-
Click	<i>mouseIndex</i>	0-3	-
dragOver	<i>mouseIndex</i>	0-3	-
dragOut	<i>mouseIndex</i>	0-3	-
releaseOutside	<i>mouseIndex</i>	0-3	-

ActionScript

```
myButton.addEventListener("click", this, "onButtonPress");
function onButtonPress(event:Object) {
    //
}
```

“ click ”

‘ myButton ’

onButt

onPress函數。相同的代碼類型也可以用來處理其他的事件。事件控制碼中返回的Object參數（例子中名為‘event’）包含了事件的相關資訊。這些資訊作為Object參數屬性返回並與多數事件有所不同。

2.1.1.6

```
import gfx.controls.Button;
attachMovie("Button", "btnInstanceName", someDepth, {_x:33, _y:262, ...});
```

CLIK

Toggle

```
btn.toggle = true;
```

```
btn.doubleClickEnabled = true;  
btn.addEventListener("doubleClick", this, "onDoubleClick");  
function onDoubleClick(e:Object):Void {  
    //  
    !  
}
```

```
btn.autoRepeat = true;
```

2.1.2 CheckBox



10

CheckBox

CheckBox

(gfx.controls.CheckBox)

true/false

ToggleButton

Toggle

2.1.2.1

CheckBox

2.1.2.2

CLIK CheckBox

MovieClip

- **textField:** () TextField
- **focusIndicator:** () MovieClip

MovieClip

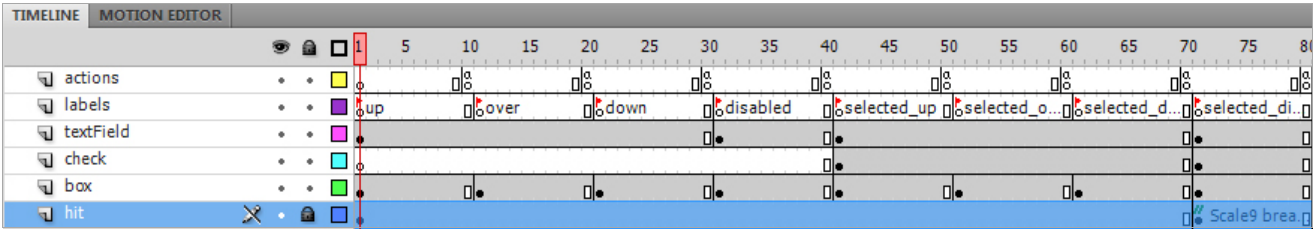
“ show ” “ hide ”

2.1.2.3

Toggle

CheckBox

- **up**
- **over**
- **down**
- **disabled**
- **selected_up**
- **selected_over**
- **selected_down**
- **selected_disabled**



11: CheckBox

CheckBox

Button

[CLIK](#)

2.1.2.4

CheckBox

disableFocus

Toggle

Data	Box	Check
disableConstraints	true	textField

	textField		disabl
	ed textField		
	textField	/	
Disabled	true		
Label			
Selected	true	CheckBox	
Visible	false		

2.1.2.5

Object

- **type:**
- **target:**

CheckBox

Show		true			
Hide		false			
focusIn					
focusOut					
Select					
	<i>Selected</i>	true			
stateChange	<i>State</i>	String	“ up ”	“ over ”	“ down ”
	<u>CLIK</u>				
Rollover	<i>mouseIndex</i>				–
		0-3			
Rollout	<i>mouseIndex</i>				
		0-3			
Press	<i>mouseIndex</i>				–
		0-3			
doubleClick		<i>doubleClickEnabled</i>	true		
	<i>mouseIndex</i>				–
		0-3			



2.1.3.1

Label

2.1.3.2

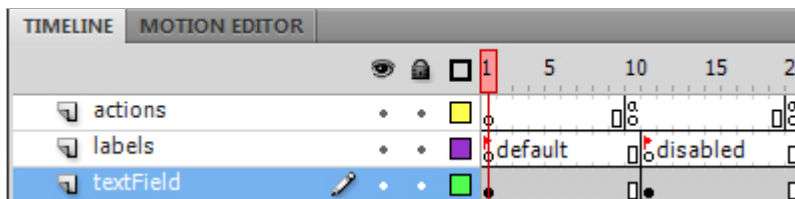
CLIK Label MovieClip

- **textField:** TextField Label

2.1.3.3

CLIK Label

- **default** enabled
- **disabled**



13: Label

2.1.3.4

Label

text	
visible	false
disabled	true

2.1.3.5

Object

- **type:**
- **target:**

Label

Show	true				
Hide	false				
stateChange	Label				
	<i>State</i>	Label	String	“ default ”	“ disabled ”

Label

```
function onStateChange(event:Object) {  
    if (event.state == "default") {  
        //  
    }  
}
```

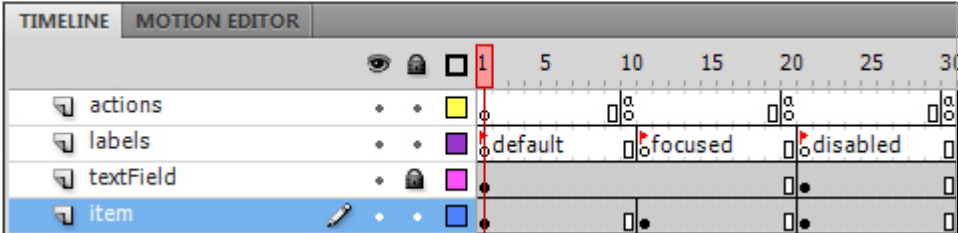
Label

- **textField:** textField

2.1.4.3 States

CLIK TextInput disabled

- **default** enabled
- **focused** textField
- **disabled**



15: TextInput

2.1.4.4

TextInput :

text	textField		
visible	false		
disabled	true		
editable	false	TextInput	
maxChars	textField		
password	true	textField	textField
defaultText	“	”	“ ”
actAsButton	“	”	“ ”
	tab “ ”		

2.1.4.5

Object

- **type:**
- **target:**

TextInput

Show	true		
Hide	false		
focusIn			
focusOut			
textChange	textField		
rollover		“	”
	<i>mouseIndex</i>		
	0 3.		
rollout		“	”
	<i>mouseIndex:</i>		
		0 3.	

textField

```
myTextInput.addEventListener("textChange", this, "onTextChange");
function onTextChange(event:Object) {
    //
}
```

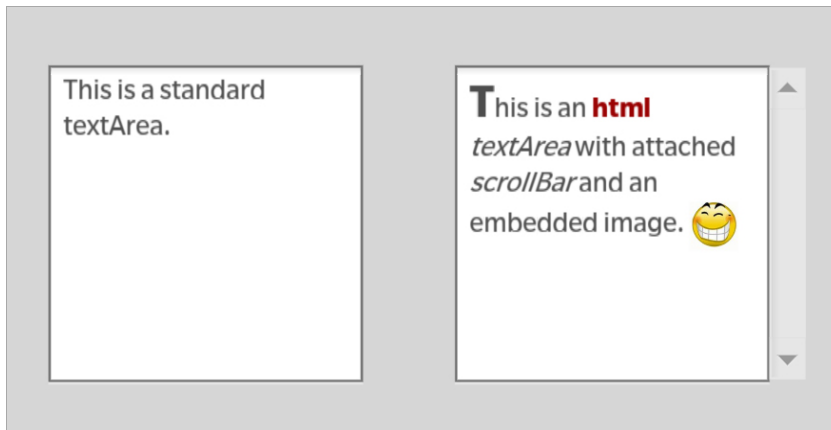
2.1.4.6

```
_global.gfxExtensions = true;
textInput.textField.noAutoSelection = true;
```

TextInput HTML textField HTML Flash 8

```
textInput.htmlText = "<b>foo</b>bar";
```


2.1.5 TextArea



16: TextArea.

TextArea (gfx.controls.TextArea) CLIK

TextInput

ScrollBar

“

”

“

”

“

“

Label TextInput TextArea

textField

textField

HTML

textField

TextArea

textField

ScrollIndicator

ScrollBar TextArea

textField

TextArea

TextArea

2.1.5.1

TextArea

textField

textField

2.1.5.2

CLIK TextArea

MovieClip

- ***textField***: TextField

TextInput TextArea disabled

-
- The screenshot shows the Timeline and Motion Editor interface. The Timeline has a horizontal axis with markers at 5, 10, 15, 20, 25, and 30 seconds. The Motion Editor shows a list of actions and their corresponding motion curves. The 'item' row is highlighted in blue. The 'actions' row has a yellow square at 10 seconds. The 'labels' row has a purple square at 10 seconds. The 'textField' row has a pink square at 10 seconds. The 'item' row has a cyan square at 10 seconds. The 'item' row also has a red vertical line at 10 seconds.

2.1.5.4

text	textField		
visible	false		
disabled	false		
editable	false	TextInput	
maxChars	.	textField	
scrollBar	CLIK ScrollBar TextArea	ScrollBar	ID
scrollPolicy	“ auto ” “ on ” ScrollBar ScrollBar	ScrollBar	“ off ”
defaultText	“ ”	“ ”	
actAsButton	“ ” “ ”	tab	“ ”

2.1.5.5

Object

- *type:*
- *target:*

TextArea

Show	true		
Hide	false		
focusIn			
focusOut			
textChange	textField		
Scroll			
rollover		“	”
	<i>mouseIndex</i>		
	0 3.		
rollout		“	”
	<i>mouseIndex:</i>		
			0 3.

TextArea

```
myTextArea.addEventListener("scroll", this, "onTextScroll");
function onTextScroll(event:Object) {
    //
}
```

2.1.5.6

```
_global.gfxExtensions = true;
textInput.textField.noAutoSelection = true;
```

TextArea HTML textField HTML Flash 8

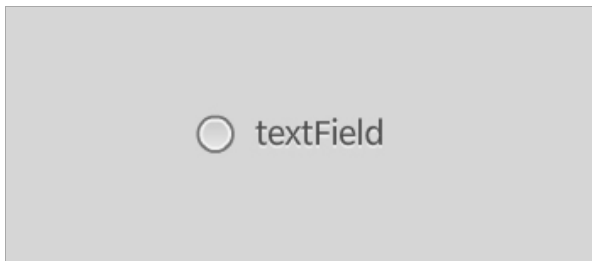
```
textInput.htmlText = "<b>foo</b>bar";
```

2.2



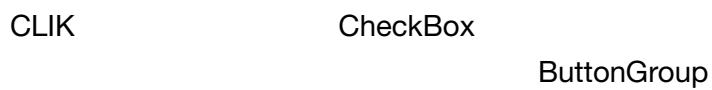
18: *Dawn of War II*

2.2.1 RadioButton



19: **RadioButton**

RadioButton
(gfx.controls.RadioButton)



2.2.1.1

2.2.1.2

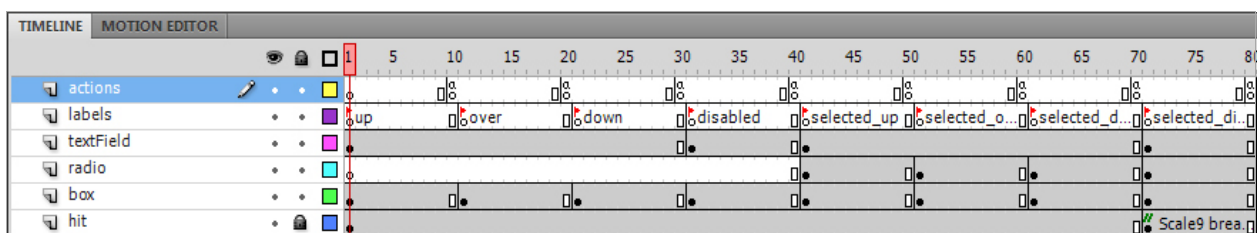
CLIK RadioButton MovieClip

- **textField** () TextField
- **focusIndicator** MovieClip MovieClip
MovieClip “ show ” “ hide ”

2.2.1.3

CheckBox

- **up**
- **over**
- **down**
- **disabled**
- **selected_up**
- **selected_over**
- **selected_down**
- **selected_disabled**



20: **RadioButton**

RadioButton Button
RadioButton CLIK

2.2.1.4

	Button	RadioButton	disable
Focus	Toggle		

Label	Button		
Visible	false		
Disabled	true		
disableConstraints	Button	textField	
	true		t
	extField		dis
	abled	textField	
		textField	/
Selected	true	RadioButton	
Data			RadioBut
	ton		
Group	ButtonGroup		RadioButto
	n	ButtonGroup	
		_root	_root

2.2.1.5

Object

- **type:**
- **target:**

RadioButton

Show	true
Hide	false
focusIn	
focusOut	
Select	
	<i>Selected</i> true
stateChange	
	<i>State</i> String “ up ” “ over ” “ down ”
	<u>CLIK</u>

rollover	<i>mouseIndex</i>	0-3	-
rollOut	<i>mouseIndex</i>	0-3	-
Press	<i>mouseIndex</i>	0-3	-
doubleClick	<i>mouseIndex</i>	<i>doubleClickEnabled</i>	true
	<i>mouseIndex</i>	0-3	-
Click	<i>mouseIndex</i>	0-3	-
dragOver	<i>mouseIndex</i>	0-3	-
dragOut	<i>mouseIndex</i>	0-3	-
releaseOutside	<i>mouseIndex</i>	0-3	-

RadioButton

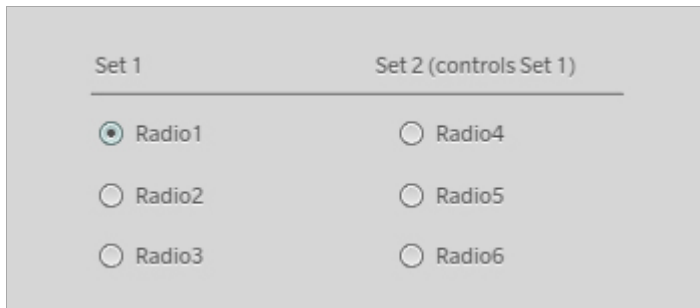
```
myRadio.addEventListener("select", this, "onRadioToggle");
function onRadioToggle(event:Object) {
    //
}
```

2.2.1.6

RadioButton

```
someMovie.radio1.group = myGroup;
someMovie.radio2.group = myGroup;
```

2.2.2 ButtonGroup



21: ButtonGroup.

CLIK ButtonGroup
(gfx.controls.ButtonGroup)

CLIK CheckBox RadioButton Butt
onGroup

2.2.2.1

RadioButton

2.2.2.2

CLIK ButtonGroup MovieClip

2.2.2.3

ButtonGroup

2.2.2.4

ButtonGroup

2.2.2.5

Object

- **type:**
- **target:**

ButtonGroup

change			
• item	CLIK	Button	
• data	AS2		
itemClick			
• item	CLIK	Button	

ButtonGroup

```
myGroup.addEventListener("change", this, "onNewSelection");
function onNewSelection(event:Object) {
    if (event.item == myRadio) {
        //
    }
}
```

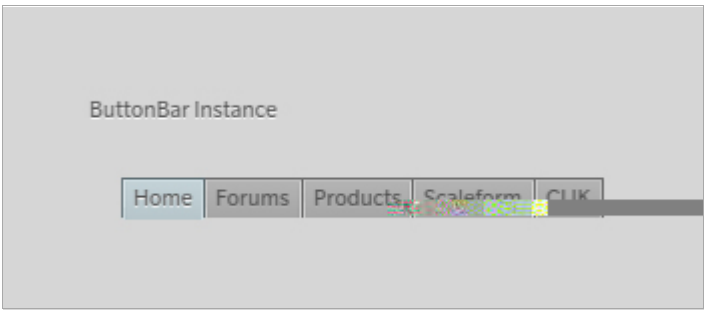
2.2.2.6

```
var selectedRadio:Button = group.selectedButton;
```

ButtonGroup

```
radioBtn1.group.addEventListener("itemClick", this, "onClick");
```

2.2.3 ButtonBar



22: ButtonBar.

ButtonBar

(gfx.controls.ButtonBar) ButtonGroup dataProvi

der dataProvider tab UI

buttonBar.dataProvider = ["item1", "item2", "item3", "item4", "item5"];

2.2.3.1

ButtonGroup

2.2.3.2

CLIK ButtonBar MovieClip

2.2.3.3

CLIK

2.2.3.4

Flash ButtonBar

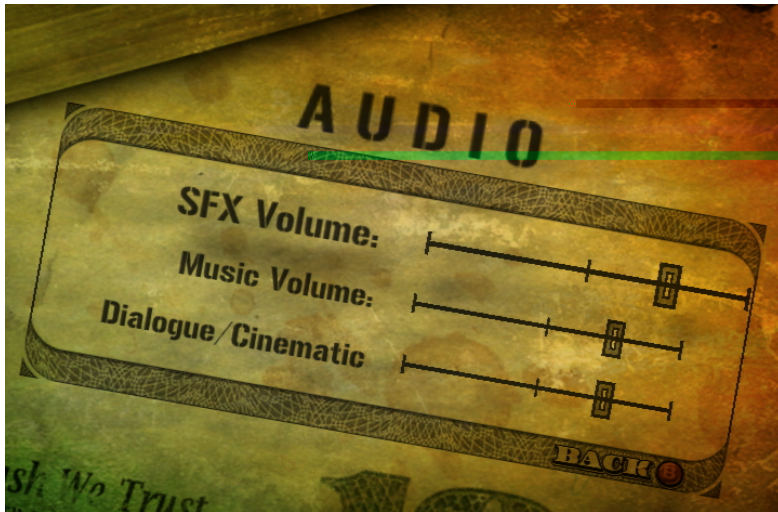
IDE

Visible	false	ButtonBar
Disabled	true	buttonBar
itemRenderer	ID	

Direction		
Spacing		<i>direction</i>
autoSize	true	

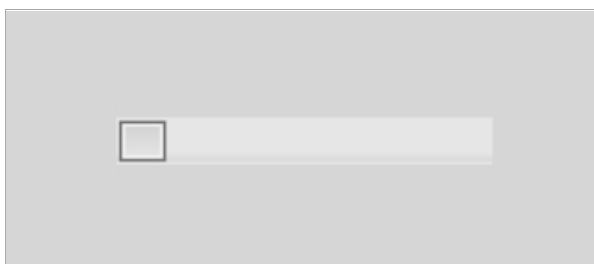
2.3

Slider ScrollIndicator ScrollBar
Slider ScrollIndicator ScrollBar
er Slid



23: *Mercenaries 2*

2.3.1 ScrollIndicator



24: **ScrollIndicator.**

CLIK ScrollIndicator
(gfx.controls.ScrollIndicator) textField
TextArea
ID

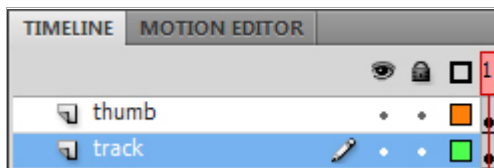
2.3.1.1

2.3.1.2

CLIK ScrollIndicator MovieClip

- **thumb**: CLIK Button
- **track** MovieClip

2.3.1.3



25: **ScrollIndicator**

2.3.1.4

scrollTarget	TextArea textField	textField ScrollIndicator
Visible	false	
Disabled	false	
offsetTop		
offsetBottom		

2.3.1.5

Object

- **type:**

- **target:**

ScrollIndicator

Show	true
Hide	false
Scroll	• <i>position</i>

```
mySI.addEventListener("scroll", this, "onTextScroll");
function onTextScroll(event:Object) {
    trace("scroll position: " + event.position);
    //
}
```

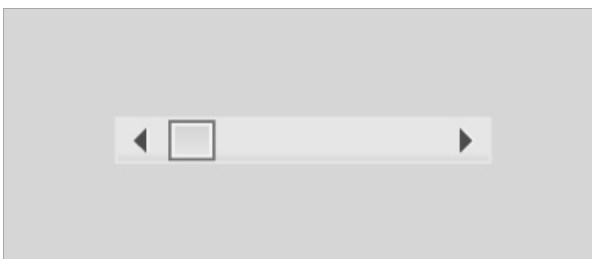
2.3.1.6

```
scrollInd.setScrollProperties(pageSize, minPos, maxPos, pageScrollSz);
scrollInd.position = currPos;
```

_rotation

```
scrollInd.direction = "horizontal";
```

2.3.2 ScrollBar



26: **ScrollBar.**

CLIK ScrollBar(gfx.controls.ScrollBar)

2.3.2.1

ScrollBar

trackMode

(Shift)

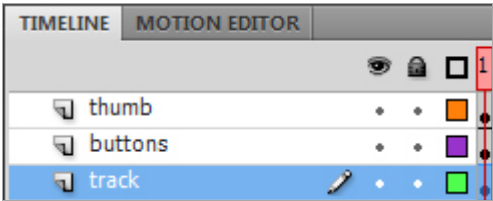
trackMode

2.3.2.2

CLIK ScrollBar MovieClip

- *upArrow* CLIK
- *downArrow*: CLIK
- *thumb*: CLIK
- *track*: CLIK

2.3.2.3



27: ScrollBar

ScrollBar ScrollIndicator

2.3.2.4

scrollTarget	TextArea	textField
trackMode	scrollPage	
	scrollToCursor	
Visible	false	
Disabled	false	
offsetTop		

offsetBottom

2.3.2.5

Object

- **type:**
- **target:**

ScrollBar

show	true
hide	false
scroll	• <i>position</i>

```
mySB.addEventListener("scroll", this, "onListScroll");
function onListScroll(event:Object) {
    trace("scroll position: " + event.position);
    //
}
```

2.3.2.6

```
scrollBar.setScrollProperties(pageSize, minPos, maxPos, pageScrollSz);
scrollBar.positon = currPos;
```

_rotation

```
scrollBar.direction = "horizontal";
```

scrollPage

1

```
scrollBar.trackScrollPageSize = pageSize;
```


2.3.3 Slider



28: Slider.

```
Slider(gfx.controls.Slider)
```

2.3.3.1

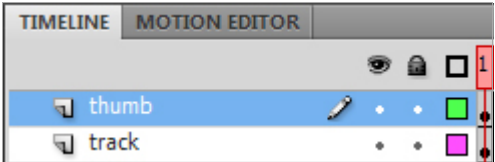
home end

2.3.3.2

CLIK Slider MovieClip

- **thumb** CLIK
- **Track** CLIK

2.3.3.3



□ 29:滑动条 Slider 时间轴

2.3.3.4

Slider

Visible	false
Disabled	false

Value	Slider	
Minimum		
Maximum		
Snapping	true	
snapInterval		false
liveDragging	true	Slider
offsetLeft		
offsetRight		

2.3.3.5

Object

- **type:**
- **target:**

Slider

show	true
hide	false
focusIn	
focusOut	
change	Slider

Slider

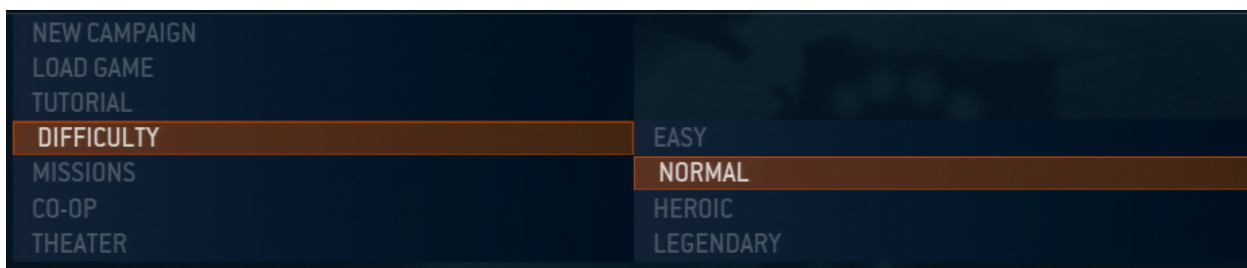
```
mySlider.addEventListener("change", this, "onValueChange");
function onValueChange(event:Object) {
    trace("slider value: " + event.target.value);
    //
}
```

2.4

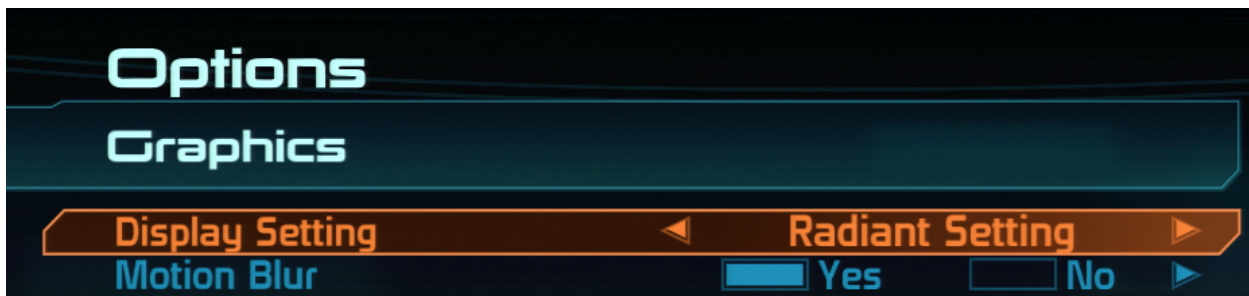
CLIK	NumericStepper	OptionStepper	ScrollingList	TileList	
DropDownMenu		NumericStepper	DataProvider		ListIt
emRenderer		ScrollingList	TileList		
NumericStepper	OptionStepper		ScrollingList		
TileList		ScrollIndicator			
ScrollBar	DropDownMenu			ScrollingList	
TileList					



30: *Mercenaries 2*



31: *Halo Wars* 'Difficulty Settings'



32: *Mass Effect* 'Display Setting'

2.4.1 NumericStepper



33: **Numeric Stepper.**

NumericStepper
(gfx.controls.NumericStepper)

2.4.1.1

NumericStepper

(End)

(Home)

2.4.1.2

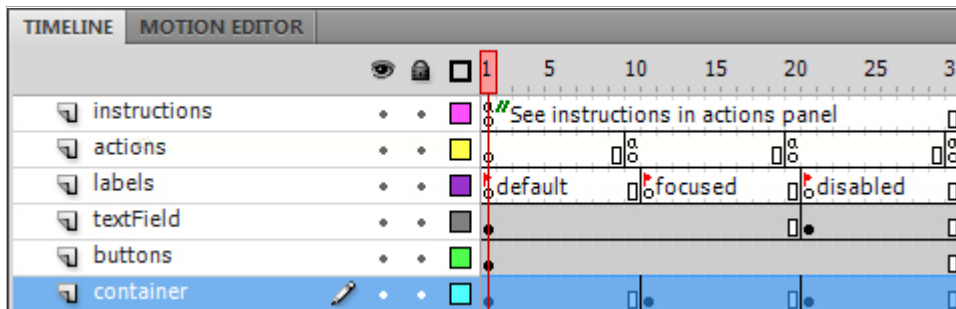
CLIK NumericStepper MovieClip

- **textField**: TextField
- **nextBtn**: CLIK Button
- **prevBtn**: CLIK Button

2.4.1.3

NumericStepper

- **default** enabled
- **focused** `textField`
- **disabled**



34: NumericStepper

2.4.1.4

NumericStepper

MovieClip

visible	false
disabled	false
value	NumericStepper.
minimum	NumericStepper
maximum	NumericStepper

2.4.1.5

Object

- ***type:***
- ***target:***

NumericStepper

Show		true		
Hide		false		
Change	NumericStepper			
stateChange	NumericStepper			
	<i>state</i>	String	“default”	“focused”
	“disabled”			

NumericStepper

```
myNS.addEventListener("change", this, "onValueChange");
function onValueChange(event:Object) {
    trace("ns value: " + event.target.value);
    //
}
```

2.4.1.6

/

```
ns.stepSize = 0.5;
```

```
ns.labelFunction = function(value:Number) {
    switch(value) {
        case 1:
            return value + "st";
        default:
            return value + "th";
    }
}
```

2.4.2 OptionStepper



35: OptionStepper.

OptionStepper

(gfx.controls.OptionStepper), NumericStepper

dataProvider

OptionStepper

dataPro

vider

```
optionStepper.dataProvider = ["item1", "item2", "item3", "item4"];
```

2.4.2.1

NumericStepper OptionStepper

(Home)

(End)

dataProvider

2.4.2.2

CLIK NumericStepper

MovieClip

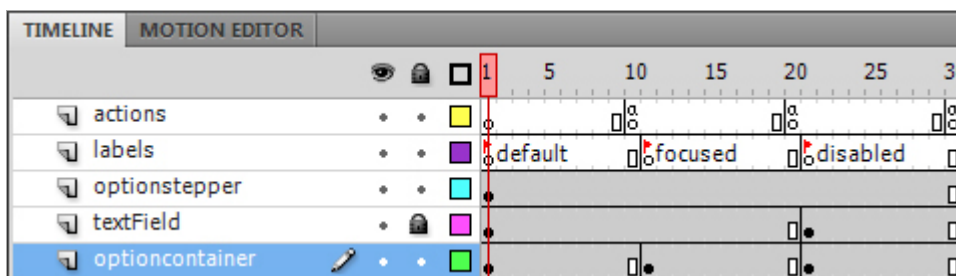
:

- **textField** TextField
- **nextBtn** CLIK Button dataProvider
- **prevBtn** CLIK Button dataProvider

2.4.2.3

OptionStepper

- **default** enabled
- **focused** textField
- **disabled**



36: OptionStepper

2.4.2.4

OptionStepper

MovieClip

Visible	false
Disabled	false

2.4.2.5

Object

- *type:*
- *target:*

OptionStepper

Show	true			
Hide	false			
Change	OptionStepper			
stateChange	OptionStepper			
	• state	String	"default"	"focused"
	"disabled"			

OptionStepper

```
myOS.addListener("change", this, "onValueChange");
function onValueChange(event:Object) {
    trace("os value: " + event.target.selectedItem);
    //
}
```

2.4.3 ListItemRenderer



图 1: ListItemRenderer.



2.4.3.1

ListItemRenderer Button
 ListItemRenderer ListItemRender
 erer

2.4.3.2

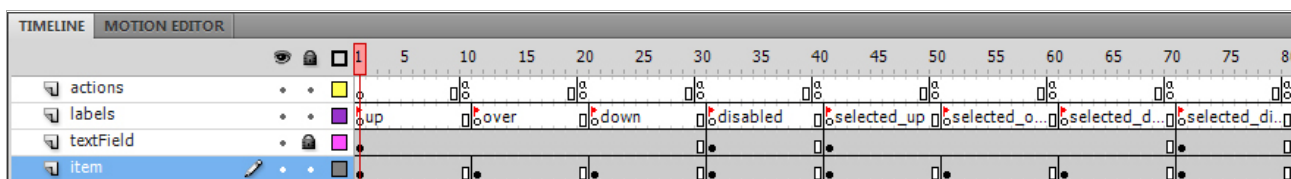
CLICK ListItemRenderer MovieClip :

- **textField** () TextField
- **focusIndicator:** () MovieClip
 MovieClip “ show ”
 “ hide ”

2.4.3.3

ListItemRenderer *selected*

- *up*
- *over*
- *down*
- *disabled*
- *selected_up*
- *selected_over*
- *selected_down*
- *selected_disabled*



38: ListItemRenderer

ListItemRenderer Button
 ListItemRenderer CLICK

2.4.3.4

ListItemRenderer

Label	ListItemRenderer
Visible	false
Disabled	true

2.4.3.5

Object

- **type:**
- **target:**

ListItemRenderer

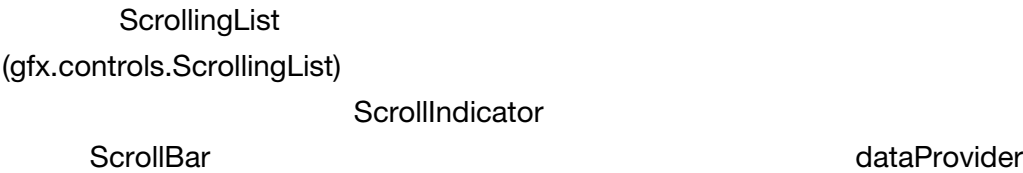
Show	true			
Hide	false			
focusIn				
focusOut				
Select				
	Selected	true		
stateChange	State	String	“ up ”	“ over ” “ down ”
	<u>CLIK</u>			
Rollover	mouseIndex	-		
		0-3		
Rollout	mouseIndex			
		0-3		
Press	mouseIndex	-		
		0-3		
doubleClick		doubleClickEnabled	true	
	mouseIndex	-		

		0-3	
Click	<i>mouseIndex</i>	-	
		0-3	
dragOver	<i>mouseIndex</i>	-	
		0-3	
dragOut	<i>mouseIndex</i>	-	
		0-3	
releaseOutside	<i>mouseIndex</i>	-	
		0-3	

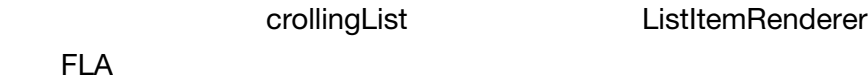
2.4.4 ScrollingList



39: ScrollingList.



```
scrollingList.dataProvider = ["item1", "item2", "item3", "item4"];
```



ormal ”

ScrollingList

ScrollingList

ScrollingList *wrapping*

wrapping

“n

wrapping

“ wrap ”

wrapping

“ stick ”

(Page Up) (Page

down)

(Home)

(End)

ScrollBar

ScrollingList

ScrollBar

Pad

CLIK

UI

D-

2.4.4.3

ScrollingList

ScrollingList

ScrollingList

- **default** enable
- **focused**
- **disabled**



2.4.4.4

ScrollingList	MovieClip
Visible	false
Disabled	true
itemRenderer	ListItemRenderer <i>rendererInstanceName</i>
rendererInstanceName	ScrollingList ‘ r ’ ‘ r1 ’ ‘ r2 ’ ‘ r3 ’ 1
Scrollbar	ScrollBar ScrollingList ScrollingList
Margin	<i>rendererIn stanceName</i>
rowHeight	<i>rendererInstanceName</i>
paddingTop	<i>rendererInstanceName</i>
paddingBottom	<i>rendererInstanceName</i>
paddingLeft	<i>rendererInstanceName</i>
paddingRight	<i>rendererInstanceName</i>
thumbOffsetTop	
thumbOffsetBottom	
thumbSizeFactor	1.0 1.0

2.4.4.5

Object

- **type:**
- **target:**

ScrollingList

Show	true		
Hide	false		
focusIn			
focusOut			
Change			
	• <i>index</i>	0	1.
itemPress			
	• <i>renderer</i>	CLIK	
	• <i>item</i>		dataProvider
	AS2		
	• <i>index</i>	dataProvider	
		0	1
	• <i>mouseIndex</i>		
		0 3	
itemClick	A list item has been clicked.		
	• <i>renderer</i>	CLIK	
	• <i>item</i>		dataProvider
	AS2		
	• <i>index</i>	dataProvider	
		0	1
	• <i>mouseIndex</i>		
		0 3	
itemDoubleClick			
	• <i>renderer</i>	CLIK	
	• <i>item</i>		dataProvider
	AS2		
	• <i>index</i>	dataProvider	
		0	1

	• <i>mouseIndex</i>	0	3	
itemRollOver				
	• <i>renderer</i> • <i>item</i> - AS2 • <i>index</i> • <i>mouseIndex</i>	CLIK		dataProvider
		dataProvider	0	1
			0	3
itemRollOut				
	• <i>renderer</i> • <i>item</i> - AS2 • <i>index</i> • <i>mouseIndex</i>	CLIK		dataProvider
		dataProvider	0	1
			0	3

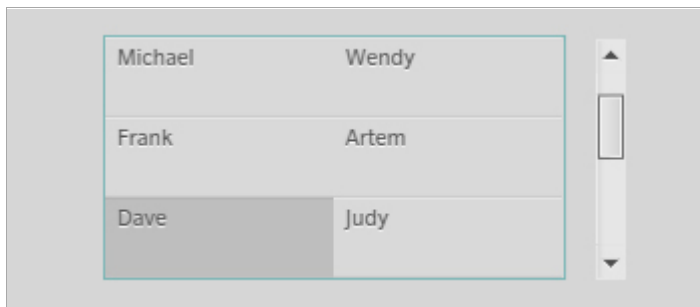
```

myList.addEventListener("itemClick", this, "onItemClicked");
function onItemClicked(event:Object) {
    trace("e135 04r4#(as Tj)/CS1 csd"0 0.439 0.753 scn1.2035 0 Td (event)Tj /CS0 cs 0

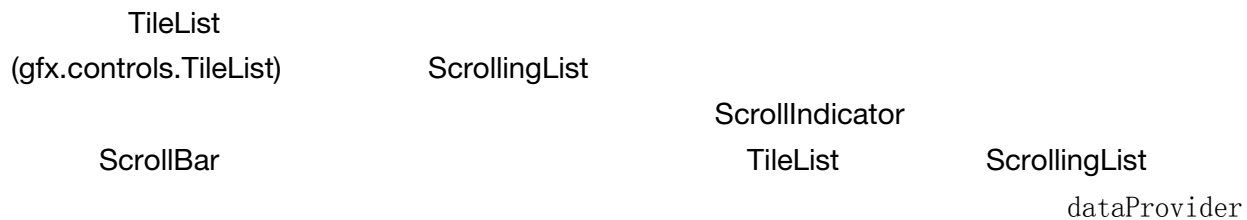
```

```
//
//      labelFunction
list.labelFunction = function(itemObj:Object):String {
    //
}
list.dataProvider = [{p1: "foo", p2: 1}, {p1: "bar", p2: 2}];
```

2.4.5 TileList



41: TileList.



```
tileList.dataProvider = ["item1", "item2", "item3", "item4", "item5"];
```



FLA

2.4.5.1



(Page Up)

(Page down)

(End)

TileList

ScrollBar

(Home)

ScrollBar

2.4.5.2

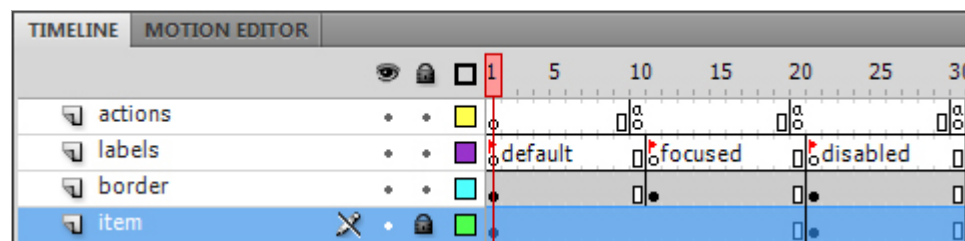
TileList

TileList

2.4.5.3

TileList

- **default** enable
- **focused**
- **disabled**



42:

TileList

2.4.5.4

TileList

MovieClip

Visible	false
Disabled	true
itemRenderer	ListItemRenderer rendererInstanceName
rendererInstanceName	TileList ' r ' ' r1 ' ' r2 ' ' r3 '

		1
scrollbar	ScrollBar	
	TileList	
	TileList	
Margin		<i>rendererIn</i>
	<i>stanceName</i>	
rowHeight		<i>rendererInstanceName</i>
columnWidth		<i>rendererInstanceName</i>
externalColumnCount	<i>rendererInstanceName</i>	
	TileList	
Direction		

2.4.5.5

Object

- *type:*
 - *target:*
- TileList

Show	true
Hide	false
focusIn	
focusOut	
Change	
• <i>index</i>	0 1.
itemPress	
• <i>renderer</i> • <i>item</i>	CLIK dataProvider
AS2	
• <i>index</i>	dataProvider 0 1
• <i>mouseIndex</i>	0 3

itemClick

- *renderer* CLIK
- *item* AS2 dataProvider
- *index* dataProvider 0 1
- *mouseIndex* 0 3

itemDoubleClick

- *renderer* CLIK
- *item* AS2 dataProvider
- *index* dataProvider 0 1
- *mouseIndex* 0 3

itemRollOver

- *renderer* CLIK
- *item* AS2 dataProvider
- *index* dataProvider 0 1
- *mouseIndex* 0 3

itemRollOut

- *renderer* CLIK
- *item* AS2 dataProvider
- *index* dataProvider 0 1
- *mouseIndex* 0 3

TileList

```
myList.addEventListener("focusIn", this, "onListFocused");  
function onListFocused(event:Object) {  
    trace("tile list was focused!");  
}
```

```
//
}
```

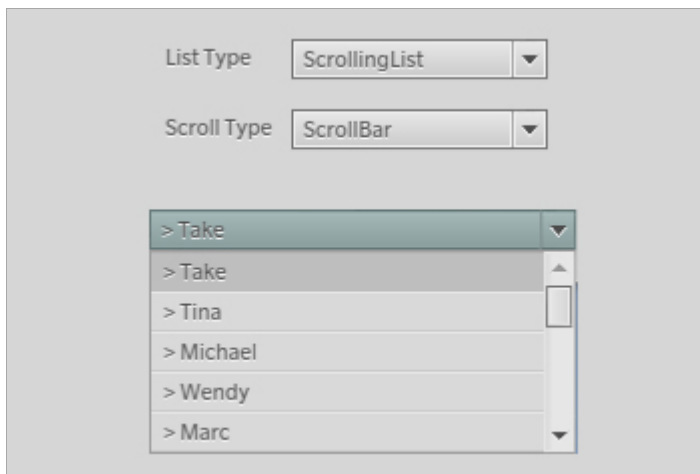
2.4.5.6

```
// ScrollingList 'label'
//
list.dataProvider = [{label: "one", data:1}, {label: "two", data:2}];

//
//
list.labelField = "name";
list.dataProvider = [{name: "one", data:1}, {name: "two", data:2}];

//
// labelFunction
list.labelFunction = function(itemObj:Object):String {
    // Logic to construct a label
}
list.dataProvider = [{p1: "foo", p2: 1}, {p1: "bar", p2: 2}];
```

2.4.6 DropdownMenu



43: DropdownMenu.

DropdownMenu
(gfx.controls.DropdownMenu)

DropDownMenu

ScrollBar

ScrollIndicator
dataProvider

ScrollingList
dataProvider
dataProvider

TileList

DropDownMenu

```
dropdownMenu.dataProvider = ["item1", "item2", "item3", "item4"];
```

DropDownMenu
DropDownMenu
ListItemsRenderer
ListItemsRenderer
FLA

ScrollIndicator
DropDownMenu
DropDownMenu
ScrollBar

2.4.6.1

DropDownMenu
(Enter)
ScrollBar
ScrollingList
TileList
DropDownMenu

2.4.6.2

DropDownMenu
MovieClip

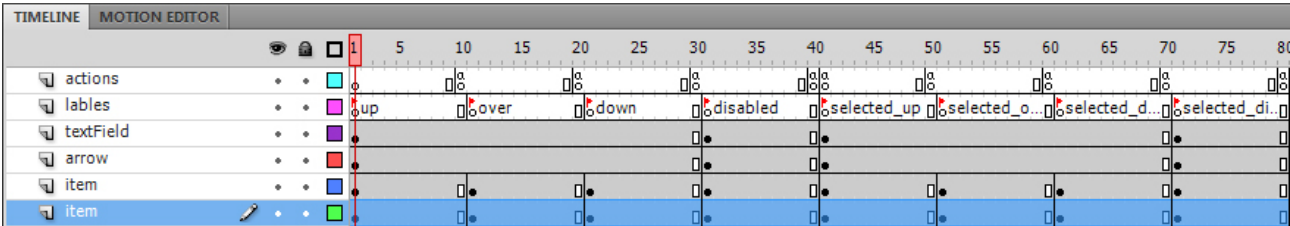
- **textField** () TextField
- **focusIndicator:** ()
MovieClip
“ show ” “ hide ”

2.4.6.3

DropDownMenu
ToggleButton
CheckBox

- **up**
- **over**
- **down**

- disabled
- selected_up
- selected_over
- selected_down
- selected_disabled



44: DropdownMenu



2.4.6.4

DropdownMenu

Visible	false
Disabled	false
Dropdown	ScrollingList
	TileList
	DropdownMenu
dropdownWidth	-
	1
itemRenderer	
scrollbar	
margin	
paddingTop	
paddingBottom	
paddingLeft	
paddingRight	
thumbOffsetTop	
thumbOffsetBottom	

Press	<i>mouseIndex</i>	0-3	
doubleClick	<i>mouseIndex</i>	0-3	<i>doubleClickEnabled</i> true
Click	<i>mouseIndex</i>	0-3	
dragOver	<i>mouseIndex</i>	0-3	
dragOut	<i>mouseIndex</i>	0-3	
releaseOutside	<i>mouseIndex</i>	0-3	-

2.4.6.6

```
dropdown.addEventListener("click", this, "onClick");
function onClick(e:Object) {
    if (e.target.isOpen) {
        // open
    }
}
```

DropDownMenu

```
//          ;
//          /          FLA
//          DropDownMenu ScrollingList
// ScrollBar .
attachMovie("DropDownMenu", "dd", this.getNextHighestDepth(),
            {dropdown: "ScrollingList"});
dd.dataProvider = ["one", "two", "three", "four", "five", "six"];
```



```

//
//
//
onEnterFrame = function() {
    dd.dropdown.scrollBar = "ScrollBar";
    onEnterFrame = null;
}

```

```

// ScrollingList          'label'
//
list.dataProvider = [{label: "one", data:1}, {label: "two", data:2}];

//
//

list.labelField = "name";
list.dataProvider = [{name: "one", data:1}, {name: "two", data:2}];
//
//      labelFunction
list.labelFunction = function(itemObj:Object):String {
    //
}
list.dataProvider = [{p1: "foo", p2: 1}, {p1: "bar", p2: 2}];

```

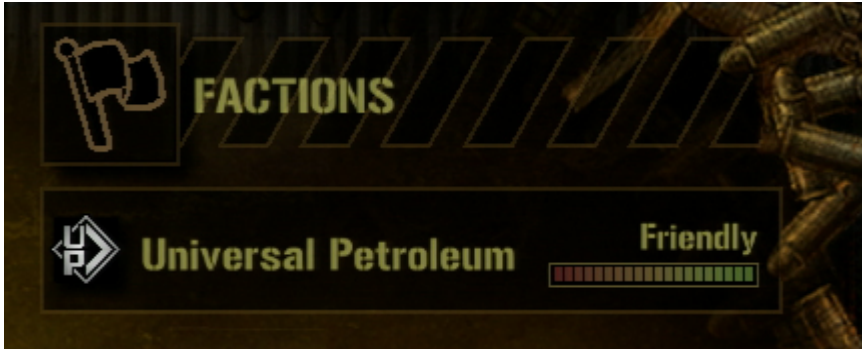
2.5

The Scaleform

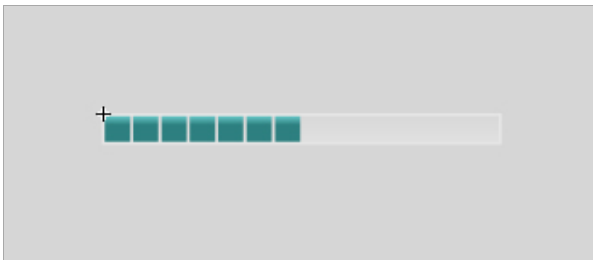
CLIK StatusIndicator

ProgressBar StatusIndicator Progr

essBar

45: *Mercenaries 2* Faction

2.5.1 StatusIndicator

46: **StatusIndicator.**

StatusIndicator component
(gfx.controls.StatusIndicator)

2.5.1.1

StatusIndicator

2.5.1.2

CLIK
StatusIndicator MovieClip
stop() goto
AndStop()

2.5.1.3

StatusIndicator

2.5.1.4

StatusIndicator MovieClip

visible	false
disabled	false
Value	
minimum	
maximum	

2.5.1.5

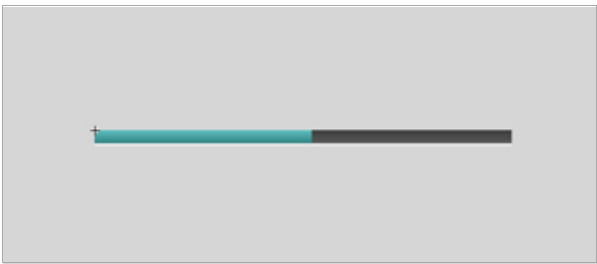
Object

- **type:**
- **target:**

StatusIndicator

Show	true
Hide	false

2.5.2 ProgressBar



47: ProgressBar.

ProgressBar
(gfx.controls.ProgressBar) StatusIndicator

bytesLoaded bytesTotal

2.5.2.1

ProgressBar

2.5.2.2

StatusIndicator CLIK StatusIndicator MovieClip
ProgressBar gotoAndStop()
ProgressBar stop()

2.5.2.3

ProgressBar

2.5.2.4

ProgressBar MovieClip

visible	false			
disabled	false			
target	ProgressBar	“	“	bytesLoaded
	bytesTotal			
mode	properties.	ProgressBar	“ manual	
	“	setProgress		“

polled "	bytesLoaded	
bytesTotal	" event "	" progress "
bytesLoaded	bytesTotal	

2.5.2.5

Object

- **type:**
- **target:**

ProgressBar

Show	true
Hide	false
progress	ProgressBar
complete	

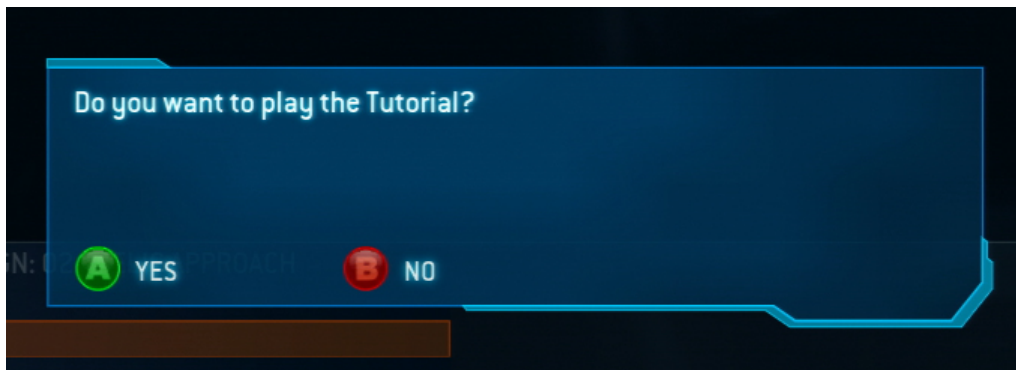
```

myProgress.addEventListener("progress", this, "onProgress");
myProgress.addEventListener("complete", this, "onProgress");
function onProgress(event:Object) {
    if (event.type == "progress") {
        //
    } else
    {
        trace("Loading complete!");
    }
}

```

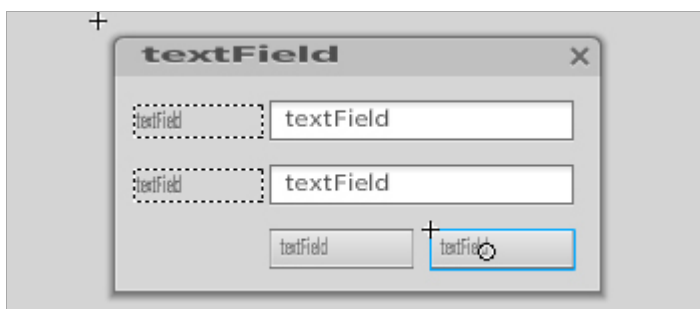
2.6

Scaleform
 CLIK
 UILoader
 ViewStack
 Dialog
 ViewStack
 UI
 Dialog
 UILoader



48: *Halo Wars* Dialog

2.6.1 Dialog



49: Dialog

CLIK
 Dialog
 (gfx.controls.Dialog)

MovieClip

MovieClip

Dialog.show()

2.6.1.1

Dialog

2.6.1.2

CLIK Dialog MovieClip :

- **closeBtn** () CLIK Button
- **cancelBtn** () CLIK Button cancel
- **submitBtn** () CLIK Button OK
- **dragBar** () MovieClip

2.6.1.3

Dialog MovieClip

2.6.1.4

Dialog MovieClip

Visible	false
Disabled	false

2.6.1.5

Object

- **type:**
- **target:**

Dialog

Show	true
Hide	false
submit	Dialog
• data	Dialog getSubmitData

Timeout			UILoa
	der	' ioError '	

2.6.2.5

Object

- *type:*
- *target:*

Ulloader

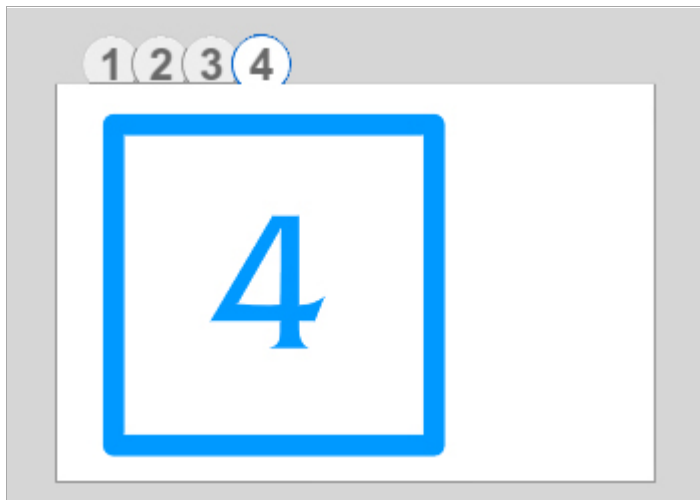
Show	true		
Hide	false		
progress		a	b
loaded	0 100		
complete			
ioError			

```

myUILoader.addEventListener("ioError", this, "onLoadingError");
function onLoadingError(event:Object) {
    displayErrorMessage("Could not load" + event.target.source);
}

```

2.6.3 ViewStack



50: **ViewStack**

CLIK ViewStack
(gfx.controls.ViewStack)

:

targetGroup	ButtonGroup	‘ change ’	ID	Ra
dioButtons		Flash IDE		ID

2.6.3.5

ViewStack

3

Scaleform

CLIK

3.1

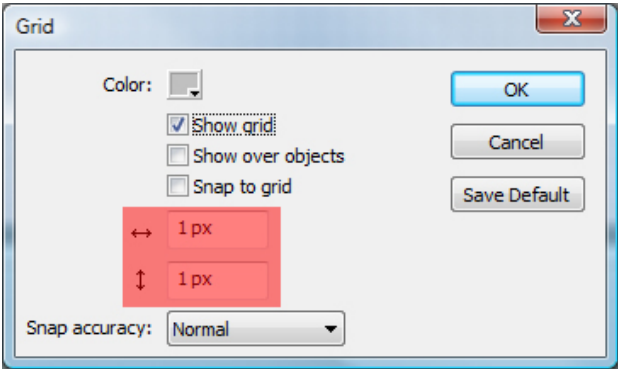
Scaleform CLIK

3.1.1

CLIK

Flash

- 1. Flash
- 2.
- 3. Flash
- 4.
- 5. 1px
- 6. OK

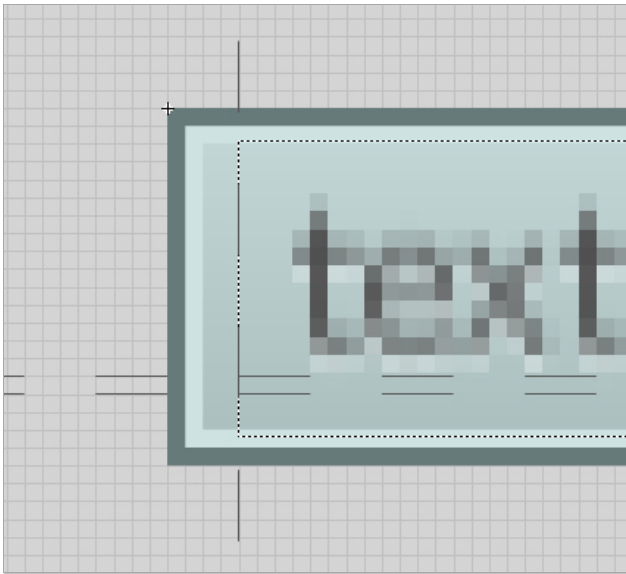


51: Grid

SWF

2

2



52:

3.1.2

Flash

Scaleform

Flash

Photoshop®

PNG

3.1.3

-

3.1.4

Flash

3.1.5

3.1.6 2

2 2

- 16x16
- 32x32
- 64x64
- 128x128
- 128x256
- 256x256
- 512x256
- 512x512

3.2

Flash

Scaleform CLIK

3.2.1

Flash

3.2.1.1 1

FLA

CLIK

FLA

1. *CLIK_Components.fl*

2. LA Paste FLA Copy F

3. FLA *Rename*

‘ Button ’

4. FLA *Properties*

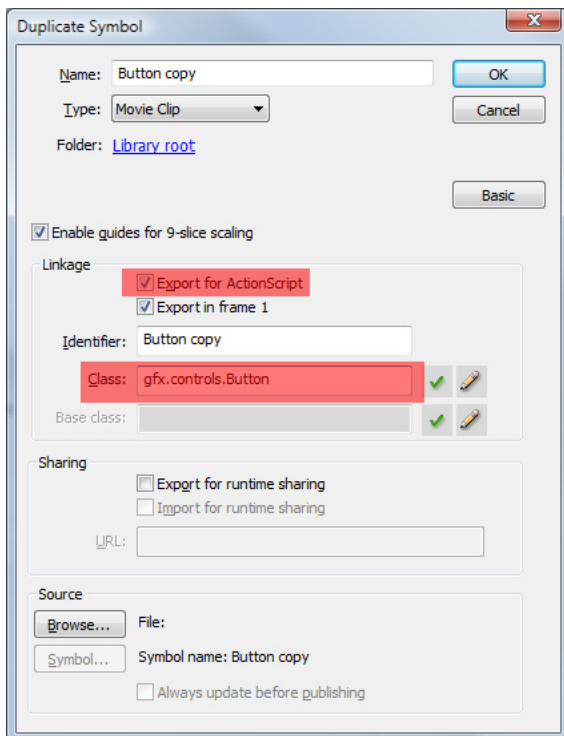
5. *Identifier* 3

6. *Paste*

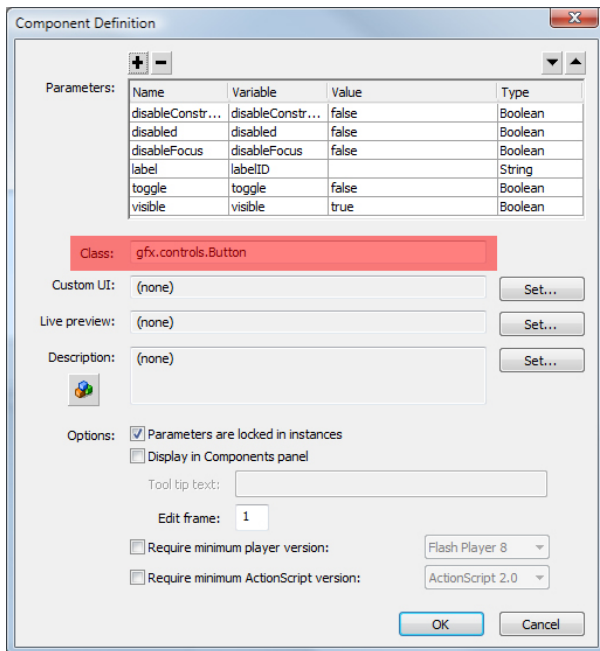
3.2.1.2

2

1. *library* *Properties*
2. *Properties* *gfx.controls.Button* *Class* (CTRL+C)
3. *Cancel*
4. *Duplicate*
5. *Duplicate Symbol* *Export* *ActionScript*
6. *Class* (CTRL+V) *textField*
7. *OK*
8. *library* *Component Definition*
9. *Class* *textField* (CTRL+V) *textField*
10. *OK*



53:



54:

3.3

Scaleform
CLIK

CLIK

CLIK

Flash

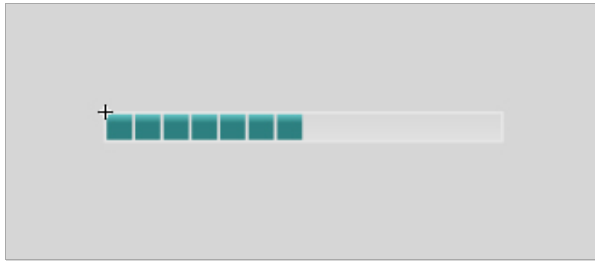
FLA

- Flash
- Flash
- Photoshop Illustrator®

Flash

[CLIK](#)

3.3.1 StatusIndicator



55: **StatusIndicator.**

StatusIndicator

indicator

track *Track*

Indicator

Photoshop



56: **StatusIndicator PSD**

16.

Remove Frames

55

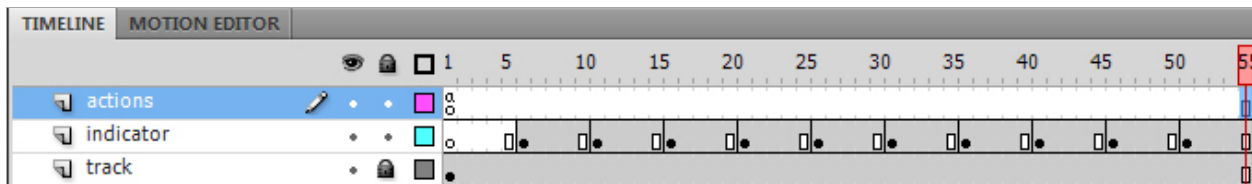
17. *indicator layer*

18. *track*

track

19. *layer 1* 'indicator'

20. *indicator track actions layer track indicator*



58:

21. *StatusIndicator*

22. *1 10*

23.

24.

3.4.2

Flash Scaleform Scaleform Scaleform textFidld Slate
Mobile Slate Mobile CJK ASCII
Slate Mobile CJK
form Scaleform textFields Scaleform 3.4.4 Scale
UI

3.4.3 textField

textFields
textField
Inspector) Embed OK (Window > Property
FLA FLA

3.4.4

Scaleform
SWF SWF Scaleform
Scaleform textField
gfxfontlib fla swf
Scaleform
fontconfig.txt Scaleform

fontconfig.txt

Scaleform

\$TITLE=Scaleform

‘ Scaleform ’

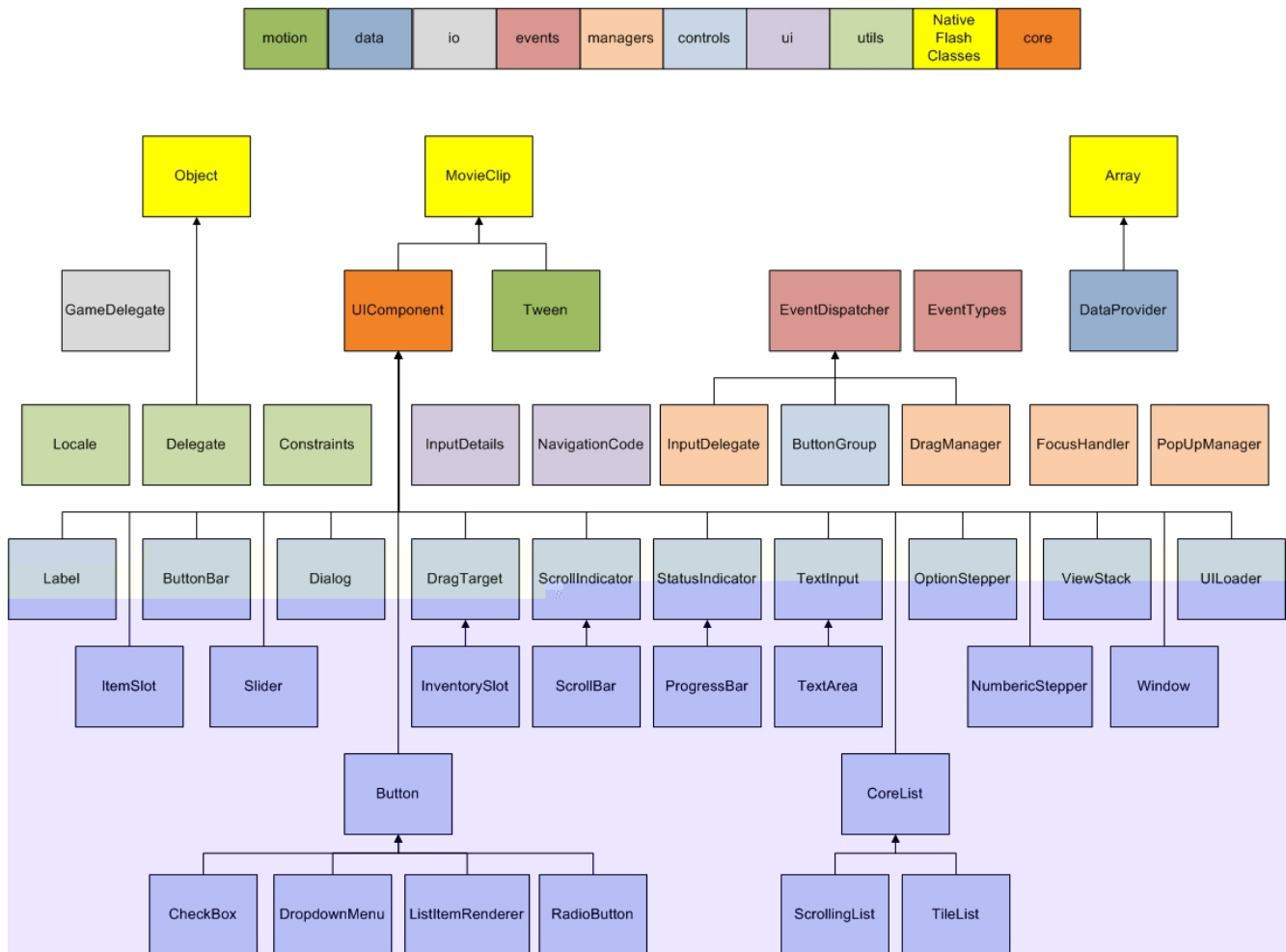
‘ \$TITLE ’

Scaleform

Scaleform

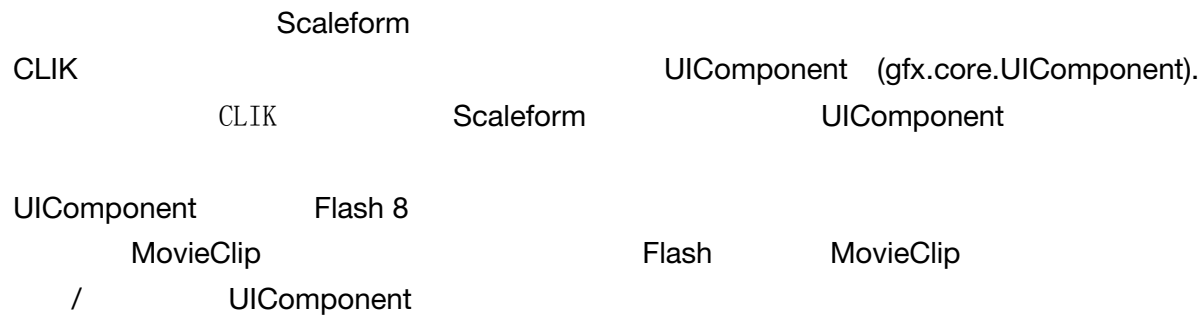


Scaleform CLIK



60: Scaleform CLIK

4.1 UI *UIComponent*



- ***disabled;***
- ***visible;***
- ***focused;***
- ***width;***
- ***height;***
- ***displayFocus:***

true

UIComponent

- ***configUI:***
- ***draw:***
- ***changeFocus:***
- ***scrollWheel:***

UIComponent EventDispatcher

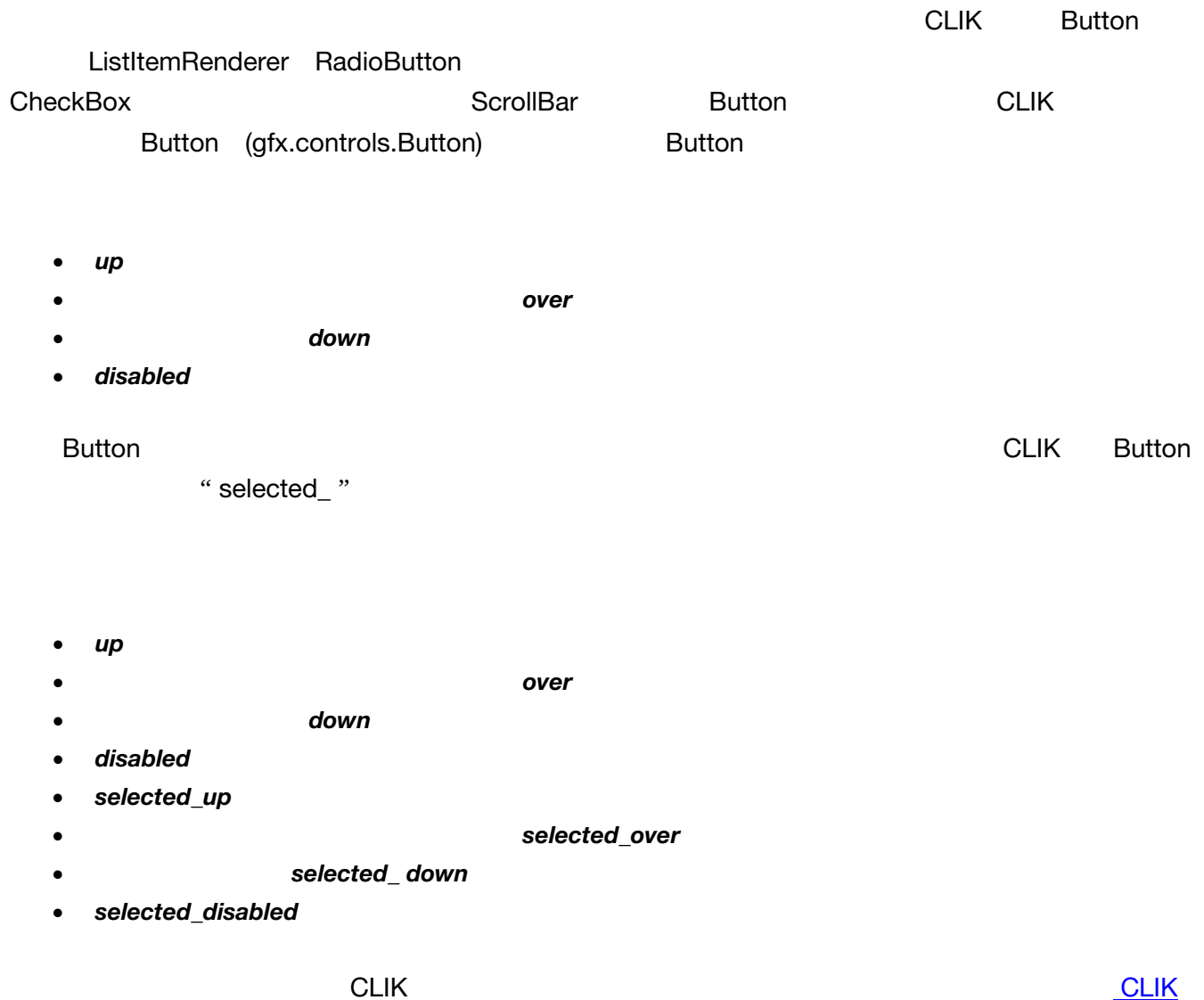
4.1.1

- ***onLoad()***
- ***MovieClip***
- ***configUI()***
- ***validateNow()***
- ***draw()***

4.2

Scaleform
CLIK

4.2.1



CLIK [CLIK](#)

CLIK getStatePrefixes()

```

private function getStatePrefixes():Array {
    return (_selected) ? ["selected_", ""] : [""];
}
  
```

CLIK " selected_ " getStatePrefixes()

stateMap

CLIK

4.2.3

- *default*
- *disabled*

4.2.4

```

                                StatusIndicator
                                Progressbar
                                50
0      10      5 50% gotoAndStop() 25 50 50%
updateValue()
                                “      ”      “
”      “      ”      “      ”      “      ”

```

4.3

```

Scaleform CLIK
model
                                “      ”
                                event
                                ActionScript 2
                                API
EventDispatcher (gfx.events.EventDispatcher)
                                EventDispatcher.initialize()
                                super()
                                UIComponent
                                CLIK
                                CLIK
                                UIComp
                                EventDispatcher
                                EventDispatcher

```

4.3.1

4.3.1.1

```

                                addEventListener()
removeEventListener()
                                String

```

```
EventTypes (gfx.events.EventTypes)
(EventTypes.SHOW " show ")
```

```
buttonInstance.addEventListener("itemClick", this, "callBack");
function callBack(eventObj:Object):Void {
    buttonInstance.removeEventListener("itemClick", this, "callBack");
}
```

“ itemClick ”

“ callback ”

CLIK

4.3.1.2

CLIK dispatchEvent()

```
dispatchEvent({type:"itemClick", item:selectedItem});
```

4.4

Scaleform

CLIK

Scaleform

ActionScript

Selection.setFocus

FocusHandler

(gfx.managers.FocusHandler)

FocusHandler

4.4.1

CLIK

Tab

true

```
Selection.setFocus(firstComponentOrMovieClip);
```


4.5

Scaleform CLIK Flash 8
 MovieClip CLIK MovieClip

4.5.1

InputDelegate class
 (gfx.managers.InputDelegate) InputDelegate InputDetails (gf
 x.ui.InputDetails) InputDelegate.readInput()
 InputDetails InputDelegate

InputDetails

- “ key ”
-
-

“ keyUp ”

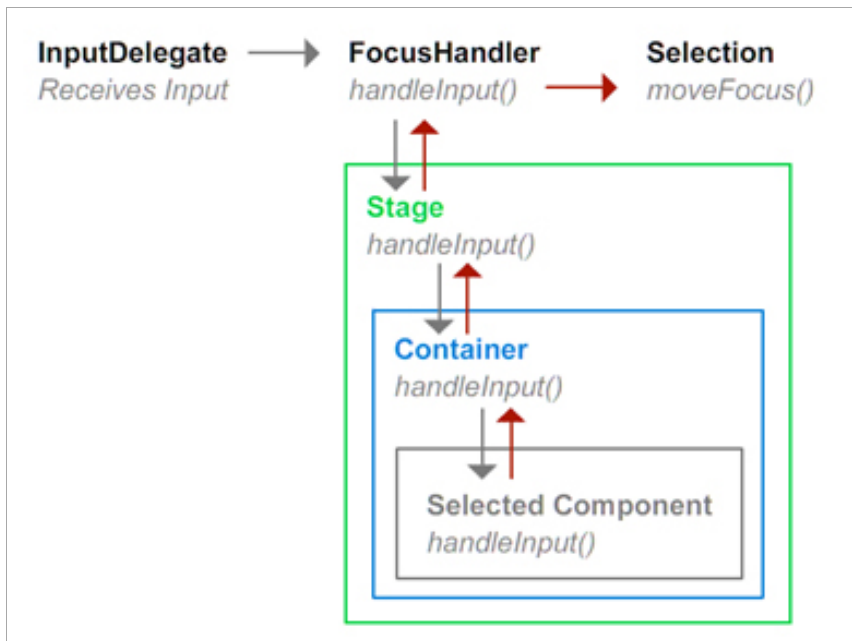
“ keyDown ”

- navEquivalent(navigation equivalent) “up” “down” “left”
- “right” NavigationCode
- (gfx.ui.NavigationCode)

FocusHandler InputDelegate

the handleInput()

InputDetails pathToFocus



62: handleInput()

FocusHandler handleInput()
 false null navEquivalent
 Scaleform

```
function handleInput(details:InputDetails, pathToFocus:Array):Boolean {
  if (details.navEquivalent == "left")
  { //    NavigationCode.LEFT
    doSomething();
    return true;
  }
  return false;
}
```

 pathToFocus true
 false

```
function handleInput(details:InputDetails, pathToFocus:Array):Boolean {
  var nextItem:MovieClip = pathToFocus.shift();
  var handled:Boolean = nextItem.handleInput(details, pathToFocus);
  if (handled) { return true; }
  //
  return false; //    true
}
```

DropDownMenu handleInput

pathToFocus

```
function handleInput(details:InputDetails, pathToFocus:Array):Boolean {
    var handled:Boolean = dropdown.handleInput(details);
    if (handled) { return true; }
    //
    return false; //          true
}
```

InputDelegate.instance
FocusHandler

```
InputDelegate.instance.addEventListener("input", this, "handleInput");
function handleInput(event:Object):Void {
    var details:InputDetails = event.details;
    if (details.value = Key.TAB) { doSomething(); }
}
```

InputDelegate

W A S D

InputDelegate

4.5.2

Nintendo

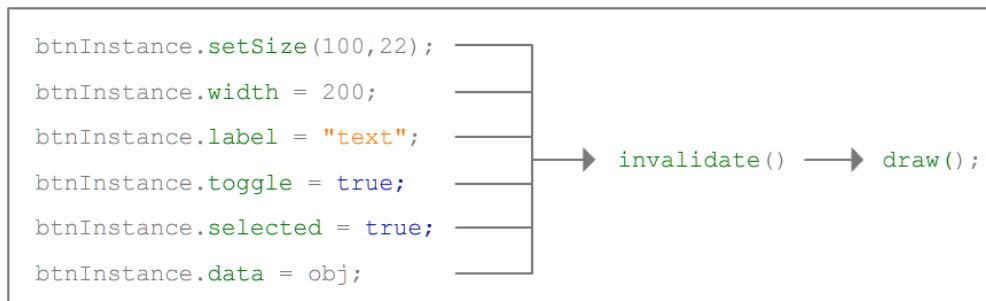
Wii™

CLIK

SWF

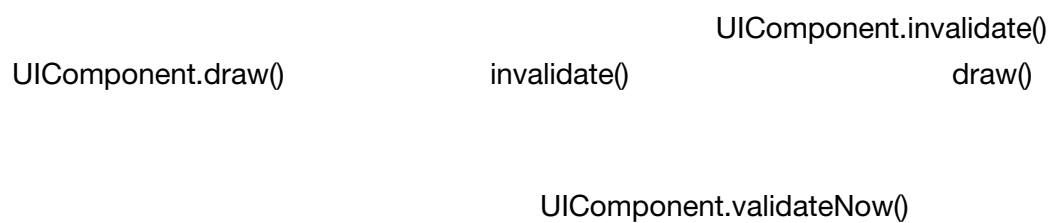
```
myButton.addEventListener("click", this, "onCursorClick");
function onCursorClick(event:Object):Void {
    switch (event.mouseIndex) {
        ...
    }
}
```

4.6



63: CLIK

4.6.1



4.7

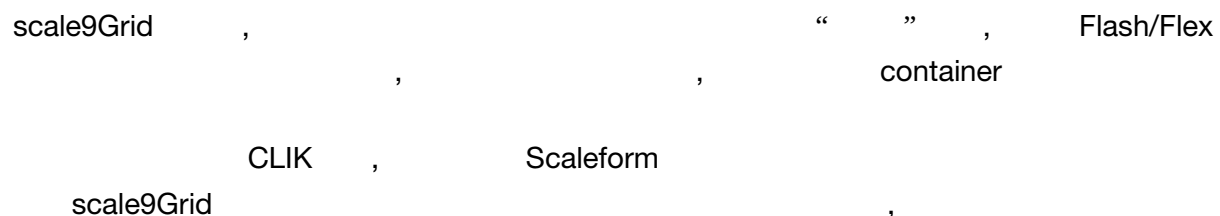
Scaleform CLIK

1.

2.

scale9grid

Flash




```

        CLIK
        ,
        container (
        ,
        Flash
        IDE
        Flash
        IDE
        MovieClips
        LivePr
        eviews
        Scaleform
        Player
        CLIK

```

4.7.1 Scale9Grid

```

        Scaleform Player
        Scale9Grid
        MovieClip
        scale9grid
        Flash
        MovieClips
        Player
        Scaleform
        MovieClip
        scale9grid
        Sca
        le9grid

```

4.7.2

```

Constraints (gfx.utils.Constraints)
        ScrollBar
        Constraints
        ScrollBar
        configUI()
        draw()
        draw()

```

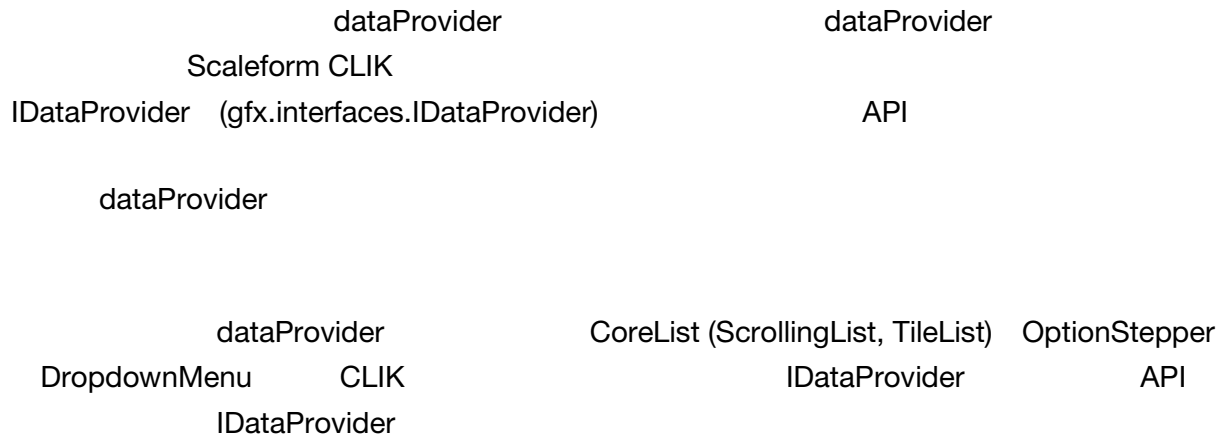
```

private function configUI():Void {
    ...
    constraints = new Constraints(this);
    //      upArrow
    constraints.addElement(downArrow, Constraints.BOTTOM);
    constraints.addElenent(track, Constraints.TOP|Constraints.BOTTOM);
    ...
}

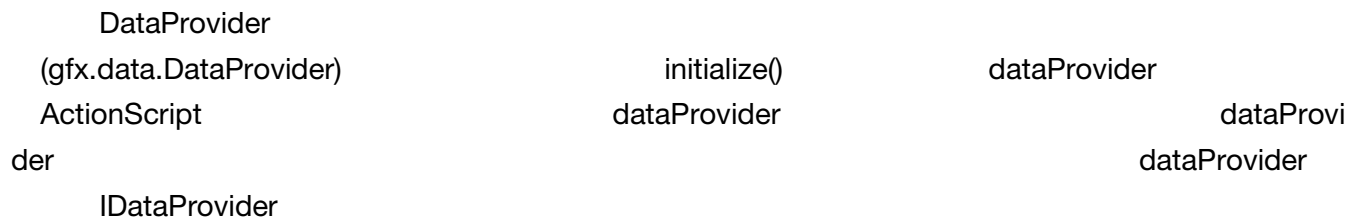
```

```
private function draw():Void {
    ...
    constraints.update(__width, __height);
    ...
}
```

4.8



4.8.1



```
myComponent.dataProvider = [ "data1",
                             4.3,
                             {label:"anObjectElement", value:6} ];
```

dataProvider

- *length*:
 - *requestItemAt*: dataProvider
OptionStepper
 - *requestItemRange*: dataProvider
ScrollingList
 - *indexOf*:
 - *invalidate*: dataProvider "dataChange"
- dataProvider

2

dataProvider ExternalInterface.call

4.9

Scaleform CLIK

Tween (gfx.motion.Tween)

Flash 8

Tween

Scaleform

Tween

easing

mx.tran

sitions.easing.*

CLIK Tween

Tween

Flash 8

MovieClip

MovieClips

Tween

CLIK

C

LIK

Tween onEnterFrame

MovieClip

Tween

Tween

MovieClip

Tween

Tween

Tween

Movi

eClip

Twee

ns

4.9.1

Tween MovieClip

tweenTo()

tweenFrom() tweenTo()

MovieClip

tweenFro

m()

Tween

MovieClip

Tween

Tween.init()

```
import mx.transitions.easing.*;
import gfx.motion.Tween;
Tween.init(); // tween MovieClip
// 1 Tween alpha
myMovieClip.tweenTo(1, {_x: 200, _alpha: 0}, Strong.easeIn);
```

Tween

Tween

onTweenComplete()

```
import mx.transitions.easing.*;
import gfx.motion.Tween;
Tween.init();
```

```
mc.onTweenComplete = function() {
    trace("Tween has finished!");
}
mc.tweenTo(5, {_rotation: 20}, Bounce.easeOut);
```

4.10

Scaleform

```
CLIK      PopUpManager  (gfx.managers.PopUpManager)
          PopUpManager
```

4.10.1

```

PopUpManager
    MovieClip
        CLIK
            ID
                createPopUp()
                MovieClips

```

```
import gfx.managers.PopUpManager;
PopUpManager.createPopUp(context, "MyToolTipLinkageID",
                          {_x: 200, _y:200}, relativeTo);
```

```

Scaleform      topmostLevel
                root
                                SWF
                                topmostLevel
                                Z
                                topmostLevel      true

```

4.11

DragManager

(gfx.managers.DragManager)

```
startDrag()           MovieClip           ID
                        stopDrag()           DragManager  InputDelegate
```

4.11.1



64: DragDemo



InventorySlot

ItemSlot

(gfx.controls.ItemSlot) InventorySlot

ItemSlot DragTarget

CLIK

ItemSI

ot

4.12

4.12.1 Delegate

Delegate

(gfx.utils.Delegate)

Scaleform

CLIK

DragManager

```
eventProvider.onMouseMove = Delegate.create(this, doDrag);
```

4.12.2 Locale

Locale (gfx.utils.Locale)

Scaleform

ActionScrip

t

Button

Label

TextInput

Locale.getTranslatedString()

Locale

5

Scaleform CLIK

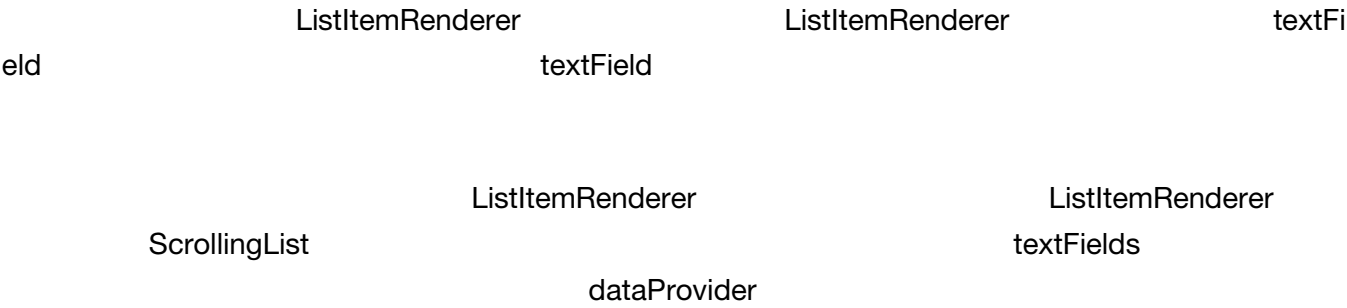
5.1

Scaleform CLIK

5.1.1 **textField ListItem Renderer**



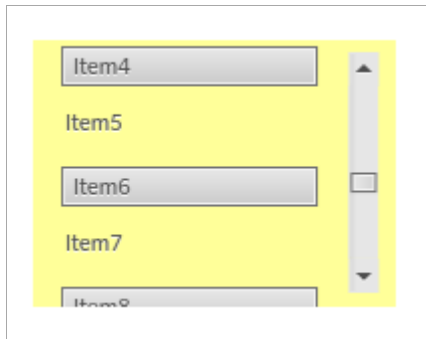
65: **ScrollingList**



```
list.dataProvider = [{fname: "Michael", lname: "Jordan"},
                     {fname: "Roger",   lname: "Federer"},
                     {fname: "Michael", lname: "Schumacher"},
                     {fname: "Tiger",   lname: "Woods"},
                     {fname: "Babe",    lname: "Ruth"},
                     {fname: "Wayne",   lname: "Gretzky"},
                     {fname: "Usain",   lname: "Bolt"}];
```



5.1.2



66:

CLIK

CLIK ScrollBar

- **actions:** ActionScript
- **scrollbar:** CLIK ScrollBar ' sb '
- **mask** MovieClip
- **content**
- **background**

' content ' y 0

ActionScript

code

```
// 133
sb.setScrollProperties(1, 0, content._height - 133);
sb.position = 0;

sb.addEventListener("scroll", this, "onScroll");
function onScroll() {
    content._y = -sb.position;
}
```

setScrollProperties

Scaleform

HUD

Scaleform

5.2

Scaleform

CLIK

Flash ActionScript 2

5.2.1 **ScrollingList TreeView**

67: TreeView

TreeView Demo

(*Resources/AS2/CLIK/demos/TreeViewDemo.fla*) ScrollingList

ListItemRenderer DataProvider

ScrollingList

-
-

“ tabbed ”

ListItemRenderer ‘ tab ’

•

ListItemRenderer

ListItemRenderer Tr

eeViewItemRenderer

TreeViewDataProvider

TreeViewConstants

Resources/AS2/CLIK/demos/c

om/

ActionScript

2

```
var root:Object =
{
    label: "System",
    nodes:[
        {label: "Applications",
          nodes:[
              {label: "CGStudio",
                nodes:[
                    {label: "Data"},
                    {label: "Samples"},
                    {label: "Config"}
                ]
              },
              {label: "Money Manager"},
              {label: "Role Call v6"},
              {label: "Super Draft",
                nodes:[
                    {label: "Templates"},
                    ...
                ]
              }
          ]
    ]
}
```

label

nodes

ScrollingList

TreeViewDataProvider

ListItemRenderer

/

/

TreeViewDataProvider

CLIK

107

-
-
-
-
-

Window

Title					
formPadding					
formType	‘ symbol ’				
	‘ swf ’	‘ symbol ’	‘ swf ’	SW	
	F				
formSource	formType		‘ symbol ’	SWF	
	formType		‘ swf ’		
allowResize					
minWidth,				0	
maxWidth,					
minHeight,					
maxHeight					

```
[Inspectable(name="formType", enumeration="symbol,swf")
private var _formType:String = "symbol";
```

Flash

_formT

type

formType

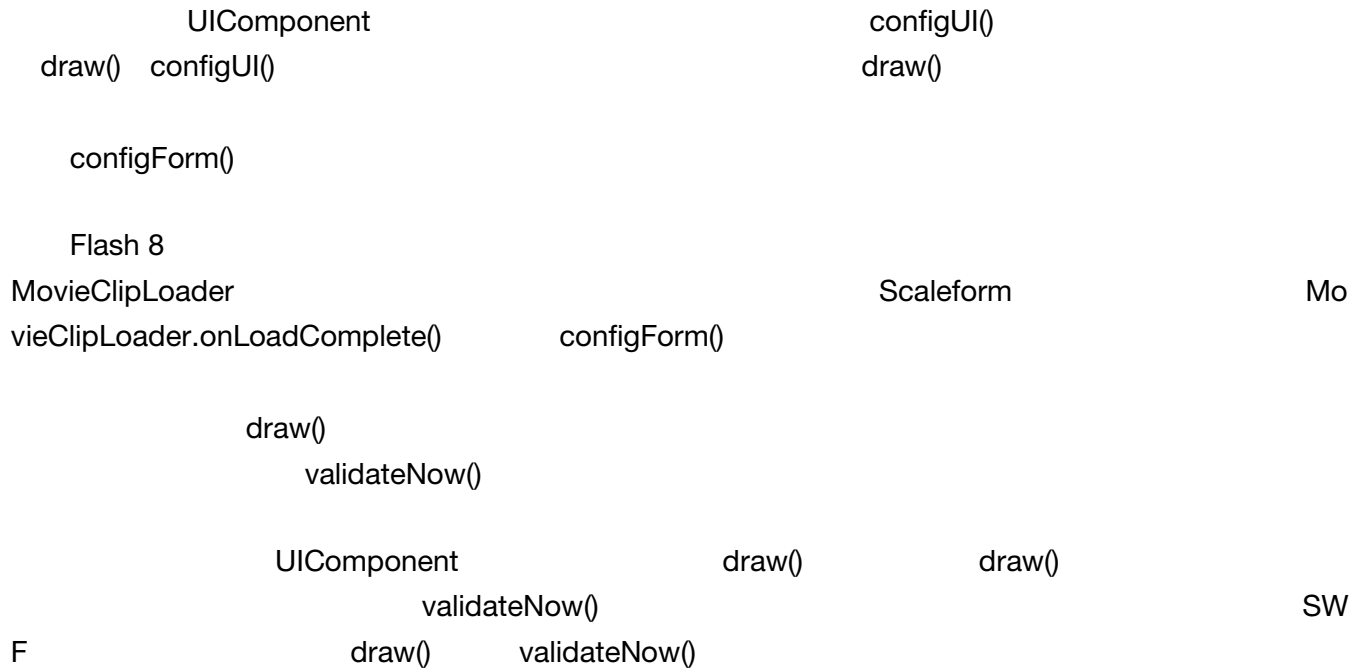
```
[Inspectable(defaultValue="Title")]
```

```

public function get title():String { return _title; }
public function set title(value:String):Void {
    _title = value;
    invalidate();
}

```

CLIK



```

private function onMouseDown() {
    var targetObj:Object = Mouse.getTopMostEntity();
    while (targetObj != null && targetObj != _root)
    {
        if (targetObj == this) {
            swapDepths(_parent.getNextHighestDepth());
            return;
        }
        targetObj = targetObj._parent;
    }
}

```

dowManager

WindowManager

z Win